INCH-POUND
MIL-C-18757/2A(OS)
25 April 1988
SUPERSEDING
MIL-C-18757/2(OS)
8 August 1974

MILITARY SPECIFICATION

CARTRIDGE CASE, STEEL, 3-INCH MARK 9 ALL MODS (50 CALIBER)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the specific requirements for the 3-inch 50-caliber Mark 9 All Mods Steel Cartridge Case, hereinafter referred to as the "Cartridge Case." (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-18757	Cartridge Cases, Steel (Other than Small Arms) General Specification for

STANDARDS

MILITARY

MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. 4387 FSC 1315

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DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND (Code Ident 10001)

255317	3-Inch/50-Caliber Cartridge Case Body Gage
375898	3-Inch Cartridge Case Mark 9 Mod 0

PUBLICATIONS

OD 47471	Method for Rating the Effects of Salt Spray Testing on Zinc Electroplated Cartridge Cases

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8	Standard Methods of Tension Testing of Metallic Materials
ASTM B 633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The complete requirements for acquiring the cartridge case described herein shall consist of MIL-C-18757, this document, and Drawing 375898.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection by a specified Government activity (see 4.4 and 6.2).

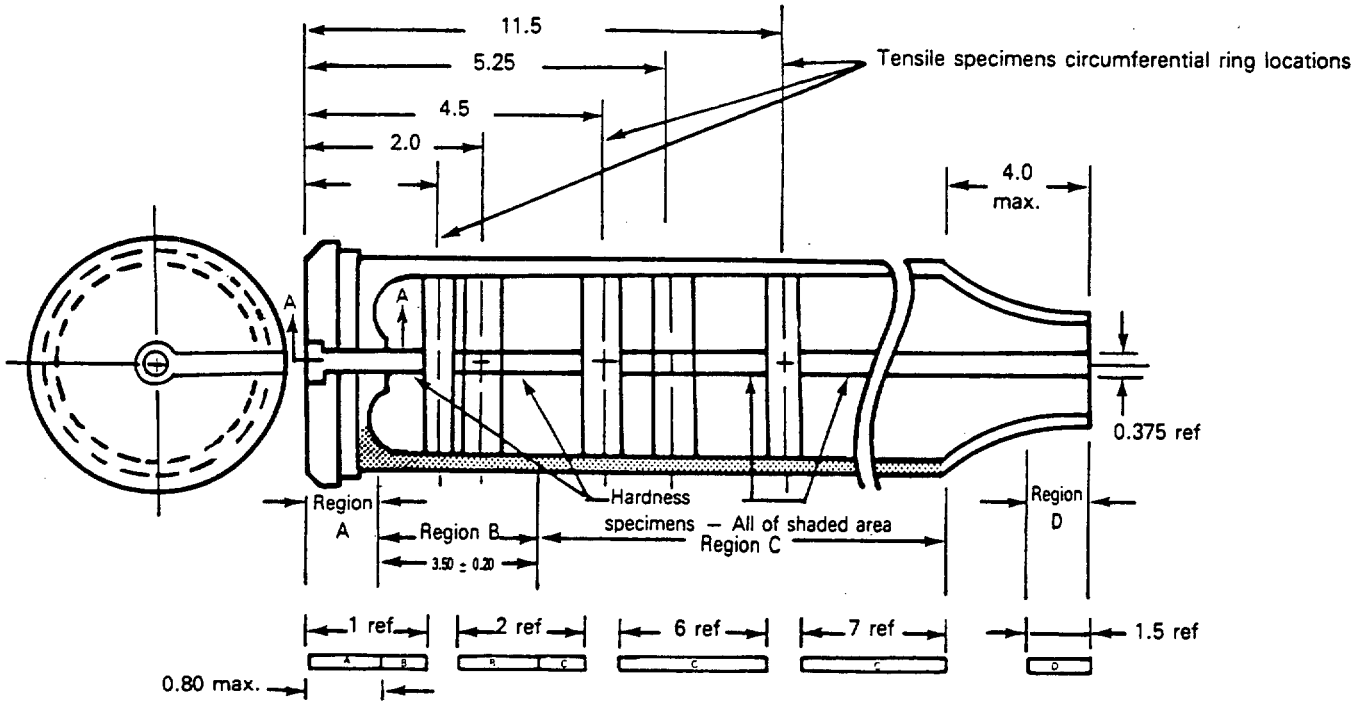
3.3 Hardness. Hardness values and determinations shall fall within the ranges specified in table I and within the test regions as shown in figure 1 (see 4.6.1). Values shown for Regions A, B, and C are advisory.

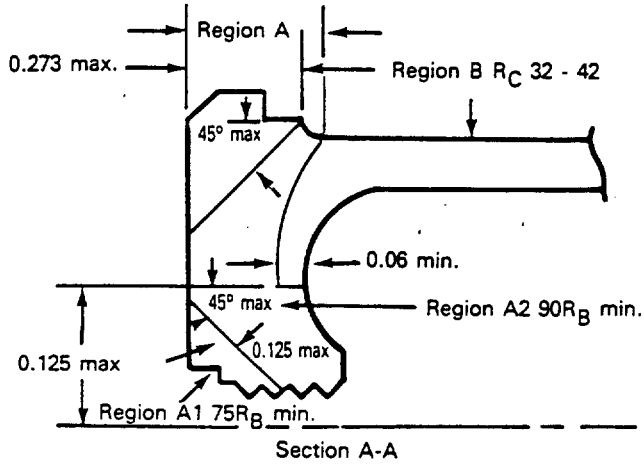
TABLE I. Hardness values.[1]

Region A_1		Region A_2		Region B		Region C		Region D	
Max[2]	Min	Max[2]	Min	Max[2]	Min	Max[2]	Min	Max[2]	Min
42C	75B	42C	90B	42C	32C	42C	94B	89B	

[1]All values given in Rockwell Scale.
[2]Paragraphs 6.5 and 6.6 of ASTM B633 do not apply.

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Tolerances: One place decimals +0.1
Dimensions: Unless otherwise specified, all dimensions are in inches

FIGURE 1. Hardness/tensile specimen locations.

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3.4 Elongation and tension. The elongation values and yield strength values shall be as specified in table II at the locations shown by figure 1 (see 4.6.2 and 4.6.3).

TABLE II. Yield strength and elongation.

Region	Yield strength (Offset-0.2 percent)	Elongation in 1 inch
B	120,000 lb/in^2	6.0 percent
C	90,000 lb/in^2	5.0 percent

3.5 Protective coating. The cartridge case protective coating shall have a rating of the black corrosion products of zinc, less than the established standard of 0.24000 (see 4.6.4).

3.6 Chambering. Each cartridge case shall fully chamber (see 4.6.6).

3.7 Ballistic test. Each cartridge case shall load, extract, and eject without difficulty when ballistically tested. Each cartridge case shall be free from cracks, tears, or splits and separations after ballistic testing (see 4.6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable paragraph(s) in the specification (see 6.2).

4.4 First article inspection. The first article shall consist of 158 cartridge cases and shall be examined and tested in accordance with table I of MIL-C-18757 and table III of this specification.

4.5 Quality conformance inspection. Fifty-eight cartridge cases shall be examined and tested in accordance with table I of MIL-C-18757 and table III of this specification.

TABLE III. Sample size for specific tests.

Type of inspection	Total sample size	Ballistic samples	Tensile hardness
First article	158	150	5[1]
Quality conformance	58	50	5[1]

[1]Exception to Note 1 to Table I, MIL-C-18757 - Hardness readings shall be obtained for all regions listed in Table I of this specification from number of samples indicated.

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4.5.1 Production lot size. Unless otherwise specified in the contract or purchase order, the production lot quantity of cartridge cases shall be 10,000 plus those required for test purposes.

4.5.2 Resinspection and retest. Reinspections and retests of any lot shall be as directed by the contracting activity.

4.6 Tests.

4.6.1 Hardness. On cases selected for hardness and tensile tests, circumferential rings for tension tests will first be cut from case per figure 1. After this, a longitudinal section with a nominal width of 0.375 inch will be cut from the remaining parts [see figure 1 (end view)] of the case to be used for obtaining hardness values for all regions specified in table I. On the circumferential ring for tension test, obtain hardness readings by taking measurements in the specimen grip area or adjacent to the grip area away from the necked down area. Record all measurements. Tests shall be in accordance with the hardness test of MIL-C-18757.

4.6.2 Elongation. Cartridge cases shall be prepared and tested in accordance with the elongation test of MIL-C-18757. Test is to be done in conjunction with tensile tests.

4.6.3 Tension. Cut and identify circumferential rings from sample cases. Maintain ring widths slightly greater than the final specimen grip width to allow for subsequent shaping operations. Prior to cutting the rings, some surface machining (both inside and outside surfaces) may be required to produce parallel specimen face surfaces (nominal thickness to be between 0.050 and 0.070 inches). Remove edge burrs and sand edges slightly to remove any possible stress raisers. Straighten specimens by passing them through a sheet metal roller, using a minimum of passes, until bowing cannot be detected with a straight edge. Machine all specimen segments to dimensions specified in ASTM E8, figure 5, subsize specimens. Several specimens may be machined at one time. Individually, sand the edges of the gage section to obtain a 0.003 - 0.005 inch taper from the end to the middle of the gage section. Smooth all edges of the necked down area. Stress relieve all segments at 610° ± 10°F for 30 ± 5 minutes and then cool to room temperature for a minimum of 1 hour. Test all tensile specimens using a head separation rate of 0.05 inch per minute. Retain original stress-strain curves. Note magnification and scale on all stress-strain curves. Measure gage dimensions as accurately as possible. Record gage cross sectional area to four decimal places.

4.6.4 Protective coating. Cartridge cases shall be prepared and tested in accordance with the protective coating tests of MIL-C-18757. The assigned plating thickness value of the interior recessed area of the cartridge case head shall be determined as follows:

4.6.4.1 Three specimen sections shall be taken from each of three cases, at approximately 120 degrees apart, and six measurements, approximately 0.1 inch apart, shall be taken on each segment (see figure 2).

4.6.4.2 The assigned plating thickness value (T) shall be calculated as follows:

T = plating thickness, any case
t_a = plating thickness, segment a
t_b = plating thickness, segment b
t_c = plating thickness, segment c
t_{a1} = plating thickness reading preceding t_{a2}
t_{a2} = minimum plating thickness reading of segment a
t_{a3} = plating thickness reading following t_{a2}

$$t_a = \frac{t_{a1} + t_{a2} + t_{a3}}{3}$$

(t_b and t_c are computed in a similar manner)

$$T = \frac{t_a + t_b + t_c}{3}$$

NOTE: (1) If t_{a2} is a measurement at location 1 or 6, use $t_{a1} = t_{a3}$ in above formula (use on any segment). (See figure 2.)

(2) Record all thickness readings.

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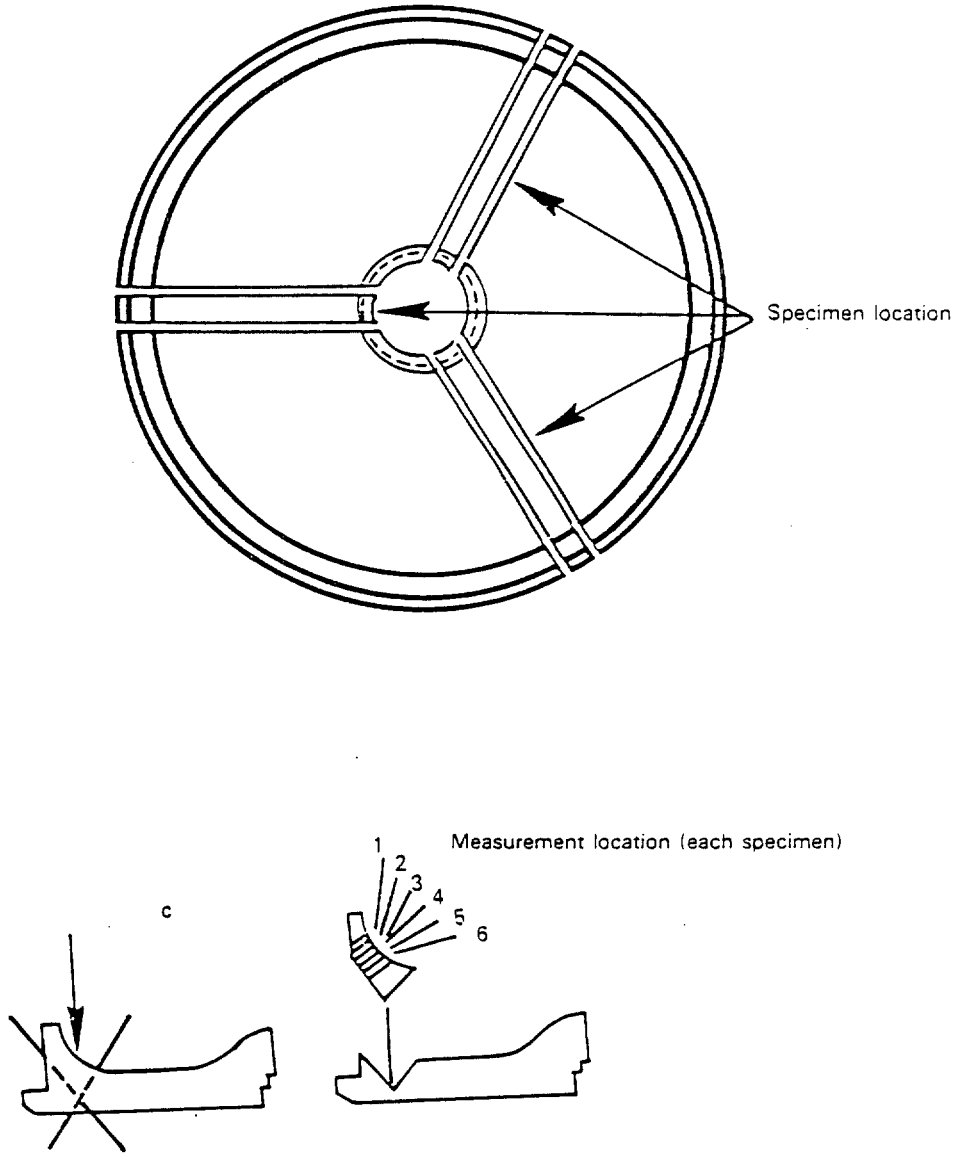


FIGURE 2. Metallographic sectioning.

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4.6.4.3 Corrosion test. The black speck corrosion rating procedure shall be as specified in OD 47471.

4.6.5 Ballistic test. The quantity of cartridge cases for ballistic testing shall be as specified in table III. The cartridges shall be loaded in accordance with drawing with an adjusted propellant charge weight to produce the pressures specified in 4.6.5.1. Firing shall be in accordance with table IV.

TABLE IV. Test firings.

Type of lot	No. of rounds	Type of rounds	Round temperature (°F)
Preproduction	70	Service	90
	30	Proof	90
	25	Service	120
	25	Service	−30
Production	40	Service	90
	10	Proof	90

4.6.5.1 Ballistic test. Cartridge cases from each lot shall be tested in the quantities as listed in table IV. Proof charge rounds shall have a test pressure between 19.0 and 21.0 TSI copper and all shall be fired in a single-shot mode. Service charge rounds shall have a test pressure between 16.5 and 17.5 TSI copper. For service pressure test, fire five rounds in single shot action with copper crusher gages to determine compliance with the pressure requirements, and fire the remainder of rounds in semi-automatic rapid fire action. Temperature of all rounds shall be as specified in table III within ±5° F.

4.6.6 Chamber gage. The 3-inch/50-caliber cartridge body gage, dwg 255317, shall be used to check compliance with 3.7.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-C-18757 and as specified herein.

5.2 Packing. Packing shall be in accordance with MIL-C-18757 and the following:

a. Nine cartridge cases shall be in each container.

b. Box dimensions shall be 14.32 × 14.32 × 23.19 inches.

c. Bottom pad dimensions shall be 22.32 × 22.32 inches corrugated, scored, 1-inch perforated score × 4.44 × 1.75 slit score × 1.75 × 4.44 × 1.75 slit score × 1.75 × 4.44 perforated score × 1 inch each way, cut four holes 3.50 × 3.50 inches where 1.75 × 1.75 inch score lines meet, notch 1 inch score eight places and at corners so it will fold up.

d. Top pad dimensions shall be 17.13 × 14.19 inches with nine holes 3.13 inch in diameter, scored across corrugation 1.30 × 14.13 × 1.50 inches.

NOTE: Tolerances for all packing dimensions shall be ±0.03 inch.

5.3 Marking. Lot number for each cartridge case lot unless otherwise specified by the contracting activity shall be in accordance with MIL-STD-1168.

5.4 Ammunition data cards. When specified in the contract or order, each production lot shall be accompanied by ammunition data cards prepared in accordance with MIL-STD-1167 (see 6.2.2).

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6. NOTES

6.1 Intended use. The 3-inch/50 caliber Mark 9, all Mods, Cartridge Cases, when assembled with their appropriate propellant primers, propellant charges, and loaded projectiles, form complete rounds which are used as service ammunition in the 3-inch/50 caliber gun.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a first article inspection is required and the Government agency that is to perform the inspection (see 3.2).

c. Whether someone other than the contractor is responsible for performance of all inspection requirements (see 4.1).

d. Inspection conditions, if other than as specified (see 4.3).

e. Production lot size, if other than as specified (see 4.5.1)

f. Lot numbering, if not to be in accordance with MIL-STD-1168 (see 5.3).

g. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475.1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.
5.4	Ammunition data card	DI-MISC-80043

(Data item descriptions related to this specification, and identified in this section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 Possible waiver. The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 158 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Ammunition
Cartridge Case, 3-inch Mark 9
Gun, 3-inch/50-caliber

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - OS
Project No. 1315-N-0936

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-18757/2A(OS)	CARTRIDGE CASE, STEEL, 3-INCH MARK 9 A11 MODS (50 CALIBER)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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